

4.1 Multimodal Foundations

Five jobs: represent, align, fuse, translate, co-learn. CLIP scores; concat fuses.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Module 2 leaves pure text. **Multimodal** systems see more than one stream: image and caption, video and audio, a PDF's pixels and its words. CMU 11-777's list is the map: **representation, alignment, fusion, translation, co-learning**. Complementary signal (the label says flammable; the photo does not) and redundant signal (lips plus audio).

If you only train on text, you cannot answer "what is in this photo?" without an image tower or a captioner. That is why Weeks 4–5 exist.

First time this method appears. Multimodal learning is more than "glue an image to a caption."

What. Five jobs (Baltrušaitis et al.): represent, align, fuse, translate, co-learn. A model can score a pair (CLIP) without decoding a sentence. **Why.** Pixels, text, and audio are different sensors. Concatenation is one fusion; it is not the only job. **Architecture.** Separate encoders per modality, then a fuse (concat, add, or cross-attention) or a score (cosine). **How.** Pick the job first. Retrieval needs a score. Captioning needs a decoder. Classification can be a linear head on a frozen encoder. **Formula.** Coordinated: $s(i, t) = \cos(f(i), g(t))$. Joint concat: $[f(i); g(t)] W$. **Tradeoffs.** + You can reuse a frozen encoder. – Concat grows width; alignment needs paired data; "multimodal" without naming the job is not a project.

1. Five jobs, not one architecture

Five jobs. CLIP scores; captioning translates; concat fuses.



Job	Question	This course
Represent	One vector, or two towers?	Joint vs coordinated
Align	Which region matches which phrase?	Attention, or one cosine on the whole pair
Fuse	When do you mix features?	Early / late / cross-attention
Translate	Map one stream into another?	Captioning (BLIP), not CLIP
Co-learn	Can a rich stream help a poor one?	Zero-shot in 4.3

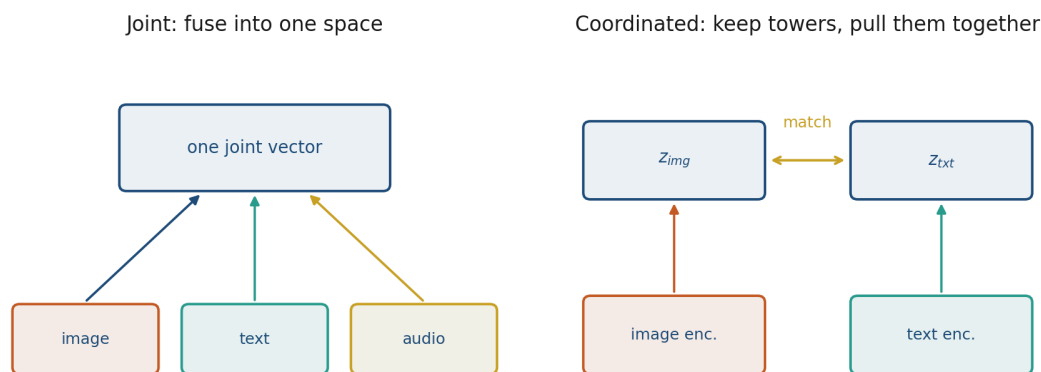
Translation is not fusion. Captioning **maps** image $\rightarrow$ text. CLIP **scores** image $\leftrightarrow$ text. If your demo must output a sentence, you need a decoder. If it must search a gallery, you need two towers and a cosine.

A detector that puts a box on “the red mug” is alignment at **region** level. A single cosine on the whole image vs the whole caption is alignment at **instance** level. Know which one your project needs.

2. Joint versus coordinated

Joint: mash the streams into **one** vector (early fusion, a concatenating MLP). Missing a modality at test time hurts.

Coordinated: keep a tower per modality, then pull the towers together with a similarity (CLIP is this; CMU 11-777 also puts CCA here). You can query with either side.



On the board: two arrows that you **add** (joint) vs two arrows that you **dot** (coordinated). CLIP never concatenates pixels with token ids during pretraining; it only scores pairs.

We do not derive Deep CCA. The slogan is enough: a coordination **loss**, not a concatenating layer.

3. Teaching this note

~**16 minutes**. Five-job table, then joint vs coordinated with the 2-D cosine example. Play Yannic Kilcher CLIP **4:40–14:40** (two towers as a coordinated space). Zero-shot details wait for 4.3; captioning waits for BLIP. Do not play a full 11-777 hour in class.

4. Worked example

Image vector $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Matching caption $t_+ = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$, other caption $t_- = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

$$\cos(v, t_+) = 0.8 / \sqrt{0.68} \approx 0.97, \quad \cos(v, t_-) = 0.$$

Coordinated retrieval ranks by cosine. Joint concat needs both streams.

pair	vectors	cosine
match	$v = [1, 0], t = [0.8, 0.2]$	0.97
mismatch	$v = [1, 0], t = [0, 1]$	0

Coordinated retrieval: rank captions by cosine. Joint fusion: $\text{concat}(v, t)$ is 4-D; you need **both** at test time, and a typed query with no image has the wrong shape.

If the image is missing, a joint MLP trained on concatenations cannot score the text alone unless you trained a special “image dropout” path. A coordinated text tower still embeds the query.

5. Where students get stuck

- Concatenating then calling it CLIP.
 - Assuming a joint model can take a text-only query at test time.
 - Mixing alignment (what matches what) with fusion (when you mix features) with translation (when you generate the other stream).
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6. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Play **4:40–14:40**. Pause when the two encoders emit vectors in one space, and when a cosine (not a concatenating MLP) is the interaction. That is coordinated representation.

The matching university course is CMU **11-777 Multimodal Machine Learning** (joint vs coordinated, then the five challenges). We do not copy those slides. Do not assign DCCA.

7. Practice

1. Your app must search photos with a typed query. Joint or coordinated? Why?
2. Give one example of alignment that is not “the whole image vs the whole caption.”
3. What breaks if a joint model never saw audio-only examples and you drop the image at test time?
4. Compute $\cos\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix}\right)$. If you fuse by concatenation instead, what is the dimension of $\text{concat}(u, v)$?
5. Two images $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and one query $q = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$. Which image ranks first by cosine?
6. Captioning versus CLIP, in one sentence each: which job from the five-job table?
7. Co-learning: English captions are plentiful, labels in another language are not. What is the shared stream that lets the rich side help the poor side?